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Validation of ReSolve HPLC Mobile Phase Recycler

An Independent Performance Evaluation of Solvent Recovery Efficiency and Chromatographic Integrity Across Multi-Sequence API Assays

Commissioned By: Antech Solutions Ltd. (Ireland)
Evaluating Body: Pharmaceutical & Molecular Biotechnology Research Centre (PMBrc), SETU
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Test System: Agilent 1260 PDA HPLC & Antech ReSolve HPLC Mobile Phase Recycler
Analytes Evaluated: Paracetamol (Acetaminophen) & Ibuprofen
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1. Executive Summary

High-Performance Liquid Chromatography (HPLC) is an indispensable analytical technique across pharmaceutical and life science production, R&D and quality control. However, standard operations generate substantial quantities of hazardous chemical waste and require vast solvent expenditure. This technical report details an independent validation study to evaluate the operational integrity, financial benefit and environmental impact of the Antech ReSolve HPLC Mobile Phase Recycler and was performed by the Pharmaceutical & Molecular Biotechnology Research Centre (PMBrc) located at SETU, Waterford.

The validation framework consisted of two consecutive analytical sequences of 96 samples, each containing a binary active pharmaceutical ingredient (API) formulation of paracetamol and ibuprofen. Using the Recycled mobile phase from Sequence 1 (Collection Stage) as the mobile phase for Sequence 2 (Evaluation Stage), the PMBrc observed no adverse impact on chromatography.

For both sequences, 94.5% of total mobile phase was recycled, with only 5.5% directed to the waste container. By reducing mobile phase consumption and subsequent hazardous waste generation a laboratory can expect:

Substantial Procurement Savings
Lowered Waste Overhead
Extended Unattended Operation

The independent validation conducted by the Pharmaceutical and Molecular Biotechnology Research Centre (PMBrc) at SETU confirms that the Antech ReSolve HPLC Mobile Phase Recycler delivers high performance without compromising analytical integrity.

2. System Overview & Technology Description

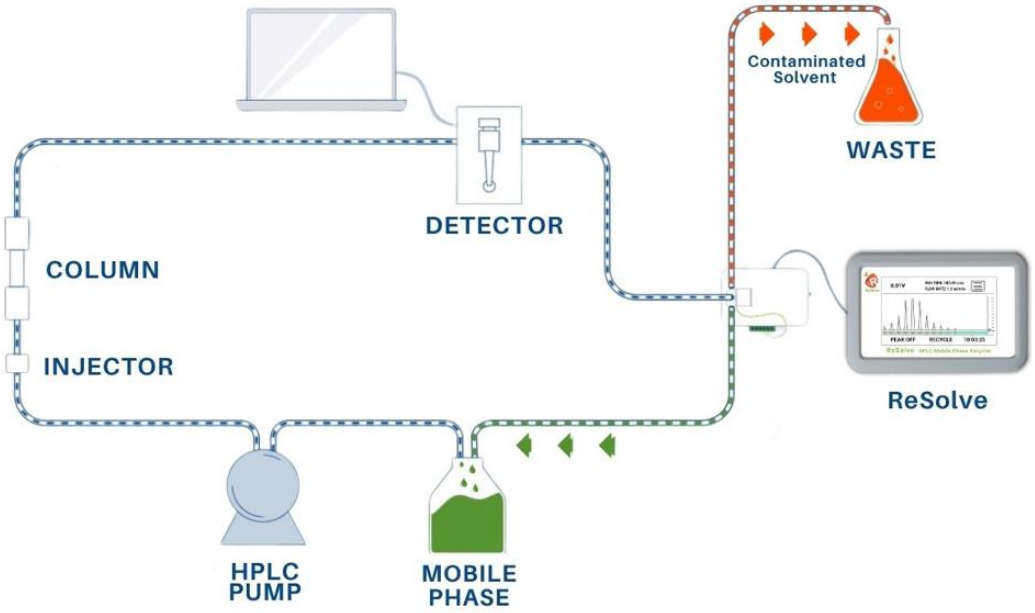


Figure 1: HPLC system incorporating the ReSolve Mobile Phase Recycler. The ReSolve recycling line is indicated with green arrows. The waste line is indicated with orange arrows.

The ReSolve Solvent Recycler functions by monitoring the analog output signal from a HPLC detector and start signal from the autosampler. During an isocratic chromatographic sequence, the mobile phase composition remains constant. When no chemical compounds are eluting (the baseline state), the fluid traveling through the detector flow cell is pure mobile phase.

The ReSolve unit integrates a high-precision and low-dead-volume switching valve directly into the waste path downstream of the detector. Guided by an intelligent and adaptive peak-sensing algorithm, the system continuously monitors detector signal voltage (V) with slope dynamics (dV/dt).

- **Recycle Mode:** When the detector signal is stable within the defined baseline threshold, the valve redirects pure, uncontaminated solvent back to the primary mobile phase reservoir.
- **Waste Mode:** Upon detecting the onset of an elution peak (dV/dt > calculated threshold), the system instantly switches the valve position, diverting the analyte peak zone directly to the hazardous waste container. The valve remains in the waste position until the trailing edge of the peak returns completely to baseline.

3. Experimental Methodology

The independent validation protocol at the PMBrc was designed to test the ReSolve instrument under demanding multi-analyte conditions to verify that no trace contaminants bypassed the diversion valve into the recycle reservoir.

3.1 Chromatographic Conditions & Sample Preparation

Chromatographic Conditions:
An isocratic HPLC methodology was developed. The mobile phase reservoir was filled with a freshly prepared batch of:

Mobile Phase: Buffer/acetonitrile/methanol (75:24.5:0.5 - % v/v/v)
Buffer: 3.9 g of Sodium dihydrogen phosphate dihydrate (25 mM) and 7.1 g of disodium hydrogen phosphate (50 mM) were dissolved in 1000 mL water and the pH adjusted to 7.0 with Orthophosphoric acid.

Column: Agilent Infinity Poroshell HPH-C8; 4.6 x 100mm; 4-Micron; Part No. 695970-706
Flow Rate: 1.0 mL/min (Isocratic)
Detection: UV at 230 nm
Injection Volume: 5 µL
Column Temperature: 35°C
HPLC System: Agilent 1260 PDA HPLC system with Antech ReSolve HPLC Mobile Phase Recycler

Sample Preparation:
A binary test mixture was prepared containing two chemically distinct and commonly encountered active ingredients with varying retention characteristics:

Analyte A: Paracetamol (acetaminophen) – Polar compound, early eluting (Retention Time (t_R) ~ 1.2 mins)
Analyte B: Ibuprofen – Non-polar compound, late eluting (t_R ~ 5.7 mins)

A single film-coated Nuromol tablet (paracetamol = 500 mg & ibuprofen = 200mg) was added to a 100 mL volumetric flask. Approximately 30 mL of mobile phase was added and the flask was thoroughly shaken and sonicated to dissolve the active ingredients. The solution was diluted to volume with mobile phase. A 1 in 50 dilution with mobile phase was performed, followed by filtration into a 1.5mL LC vial (0.2 µm Pore Size) for injection onto the HPLC system.

Standard Preparation:
A binary standard of paracetamol (99.5 µg/mL) and ibuprofen (40.6 µg/mL) was prepared in mobile phase.

3.2 Dual Sequence Experimental Design

The validation process was separated into two distinct, large sequences to rigorously assess chromatographic impacts of recycling using the ReSolve unit:

Table 1: Experimental Design.

Validation Stage	Sample Count	Mobile Phase Source	Objectives
Sequence 1 ***Collection Stage***	96 Injections 80 Samples 13 Standards 3 Mobile Phase Blanks	Freshly prepared, pristine mobile phase.	To perform initial sample analyses while simultaneously using the ReSolve unit to separate non-peak events into a dedicated recycle vessel and peak events into a dedicated waste vessel.
Sequence 2 ***Evaluation Stage***	96 Injections 80 Samples 13 Standards 3 Mobile Phase Blanks	100% recycled mobile phase recovered from Sequence 1.	To repeat the identical 96-sample sequence using the recycled solvent from Sequence 1 as mobile phase. To detect any baseline degradation, ghost peaks, peak shape change, area change or shifts in retention times.

The impact of using the recycled mobile phase from Sequence 1 (Collection Stage) as mobile phase for Sequence 2 (Evaluation Stage) was assessed.

4. Results and Discussion

4.1 Volumetric Recycling Efficiency

Across the 96 injections of both sequences, the ReSolve system accurately registered peak start and end events for both paracetamol and ibuprofen. Mobile phase volumes are summarised in Table 2.

Table 2: Recycled and Waste volumes observed during HPLC analysis.

Sequence	Total Volume of Mobile Phase used (mL)	ReSolve Recycled Volume (mL)	ReSolve Waste Volume (mL)
Sequence 1 Collection Stage 96 Injections	1150.5	1087.0 **94.5% Recycled**	63.5 **5.5%**
Sequence 2 Evaluation Stage 96 Injections	1150.6	1087.1 **94.5% Recycled**	63.5 **5.5%**

Data taken from ReSolve unit.

For both sequences, 94.5% of total mobile phase was recycled, with only 5.5% directed to the waste container. Typical activity (i.e. switching between Recycling and Waste) of the ReSolve unit for a paracetamol / Ibuprofen sample is shown in Figure 2.

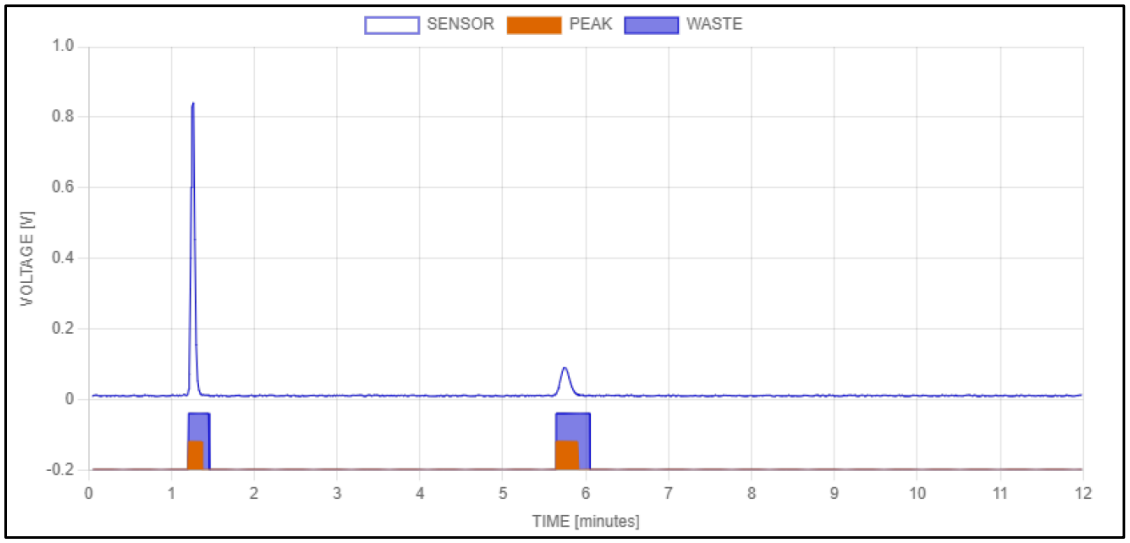


Figure 2: Typical chromatogram for a paracetamol / ibuprofen sample, showing ReSolve unit activity (Orange Square = Peak detected; Blue Square = Flow diverted to waste, incorporating a conservative safety margin. The unit remains in Waste mode for an additional 50% of Peak Duration to facilitate any extended tailing that may be present).

4.2 Chromatographic Equivalency Analysis (Sequence 1 *vs* 2)

The critical indicators of validation success lie in the analytical data of Sequence 2 (Evaluation Stage). Should using the ReSolve system have resulted in a change in mobile phase composition or accumulated contamination from multiple injections, changes in the chromatographic performance in Sequence 2 might be expected.

As shown in Table 3, comparative analysis revealed that the chromatography metrics for Sequence 1 and Sequence 2 were comparable (see APPENDIX 1 for detailed data).

Retention Times (t_R): The retention times for both paracetamol and ibuprofen matched between sequences, confirming that the recycling did not alter the organic-to-aqueous ratio of the mobile phase due to selective evaporation or composition adjustment.
Peak Area: The peak areas of the runs with the fresh and recycled mobile phases were comparable, indicting no background matrix interference or contamination which altered detector response.
Tailing Factor (T_f): Tailing for both paracetamol and ibuprofen matched between sequences confirming that chromatography was not impacted by recycling using the ReSolve system.
Symmetry Factor (*As*): Peak Symmetry for both paracetamol and ibuprofen was equivalent for both sequences, further confirming that chromatography was not impacted by recycling using the ReSolve unit.
Peak Resolution–EP (*Rs*): Peak Resolution for paracetamol and ibuprofen was also equivalent for both sequences, confirming again that chromatography was not impacted by recycling using the ReSolve unit.
Baseline Noise: No difference in baseline noise/stability was observed between Sequence 1 & 2.

Table 3: Chromatographic Metrics for Sequence 1 and Sequence 2.

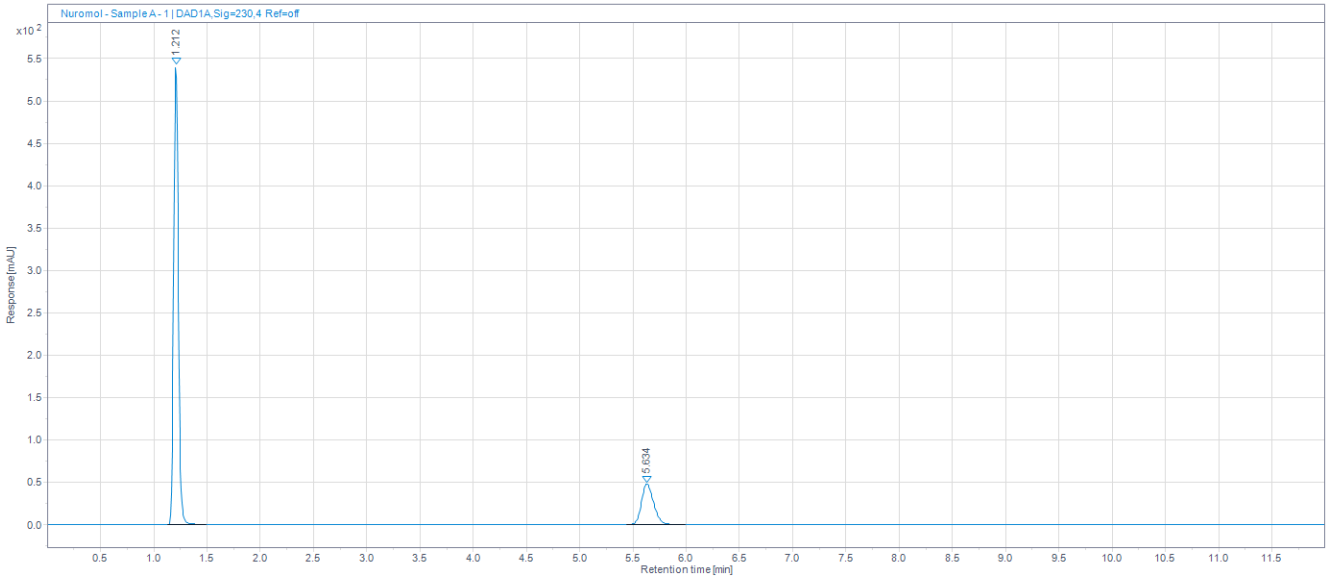
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Metric	Sequence 1 Collection Stage (Average ± sd)	Sequence 2 Evaluation Stage (Average ± sd)	% Difference
Paracetamol Retention Time; mins (Sample. n = 80)	1.21 ± 0.00	1.21 ± 0.00	0.00
Paracetamol Peak Area; mAU.s (Sample. n = 80)	1670.9 ± 4.7	1673.1 ± 5.2	+ 0.1
Paracetamol Tailing (Sample. n = 80)	1.12 ± 0.04	1.12 ± 0.05	0.00
Paracetamol Symmetry (Sample. n = 80)	0.87 ± 0.00	0.86 ± 0.01	- 1.16
Ibuprofen Retention Time; mins (Sample. n = 80)	5.65 ± 0.01	5.67 ± 0.01	+ 0.35
Ibuprofen Peak Area; mAU.s (Sample. n = 80)	375.8 ± 0.7	375.1 ± 0.8	- 0.2
Ibuprofen Tailing (Sample. n = 80)	1.21 ± 0.02	1.21 ± 0.02	0.00
Ibuprofen Symmetry (Sample. n = 80)	0.75 ± 0.00	0.75 ± 0.00	0.00
Peak Resolution EP (Sample, n = 80)	30.31 ± 0.04	30.27 ± 0.05	- 0.13

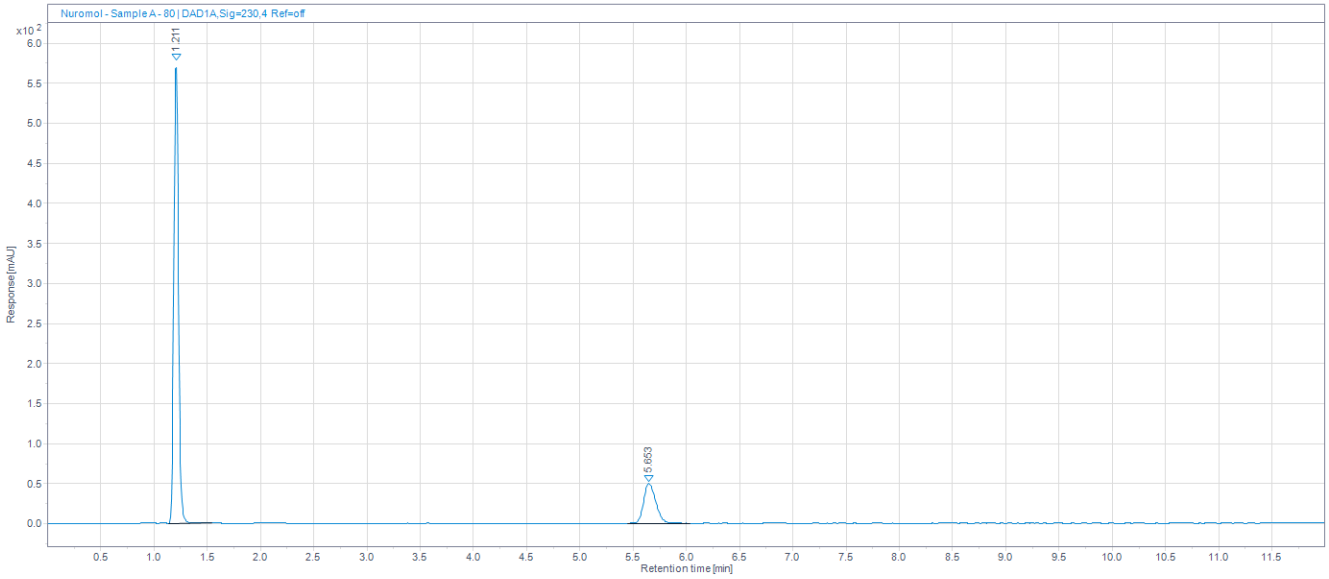
Chromatographic performance was equivalent between Sequence 1 (Collection Stage) and Sequence 2 (Evaluation Stage), with minimal differences in metrics observed between sequences. The use of the recycled mobile phase from the Collection Stage has had no significant impact on chromatographic performance in the Evaluation Stage.

Using LC/MS, the concentration of paracetamol and ibuprofen in the Recycled and Waste phases of Sequence 1 (Collection Stage) were measured. Paracetamol and ibuprofen levels of 1440 ng/mL and 530 ng/mL, respectively, were observed in the Waste phase. Paracetamol or ibuprofen was not detected in the Recycled phase.

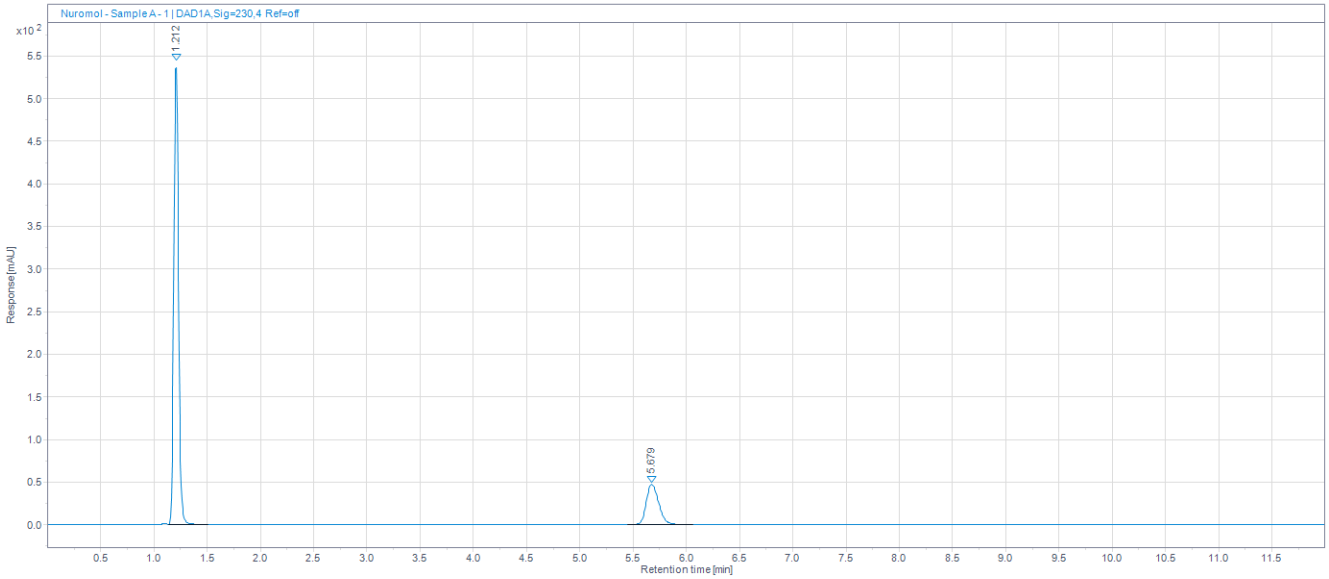
Examples of Chromatograms:



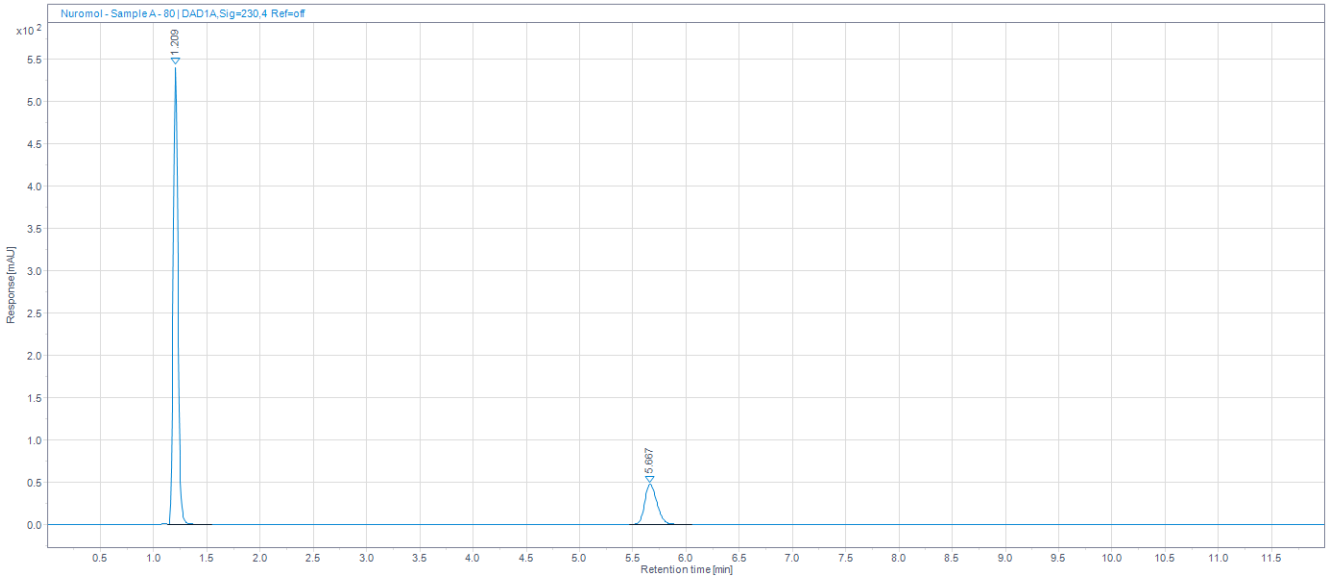
Sequence 1 (Collection Stage) – Sample Injection 1



Sequence 1 (Collection Stage) – Sample Injection 80



Sequence 2 (Evaluation Stage) – Sample Injection 1



Sequence 2 (Evaluation Stage) – Sample Injection 80

Figure 3: Typical chromatograms (Injection 1 & Injection 80) for paracetamol and ibuprofen samples for Sequence 1 (Collection Stage) and Sequence 2 (Evaluation Stage).

ReSolve
HPLC Solvent Recycler
Injection Log

Company: PMBrc
Laboratory: HPLC Lab B21
HPLC System: Agilent 1260 DAD
ReSolve Serial No: 242100001

Filename: Chromatogram_2026_05_06_07_48_06.pom
Injection Date & Time: 06-05-2026_07:48:06
User: Unknown
Flow Rate: 1.00 ml/min
Runtime: 12.0 min
Delay: 1.86
Slope: 1000
Width: 20 s
Threshold: 0 mV

Mobile Phase to Waste Events

EVENT	From (mins)	To (mins)	Volume (ml)
WASTE 1	01.21	01.46	0.25
WASTE 2	05.61	06.02	0.41
Volume to Waste			0.66
Volume Recycled			11.32
Percentage Recycled			94%

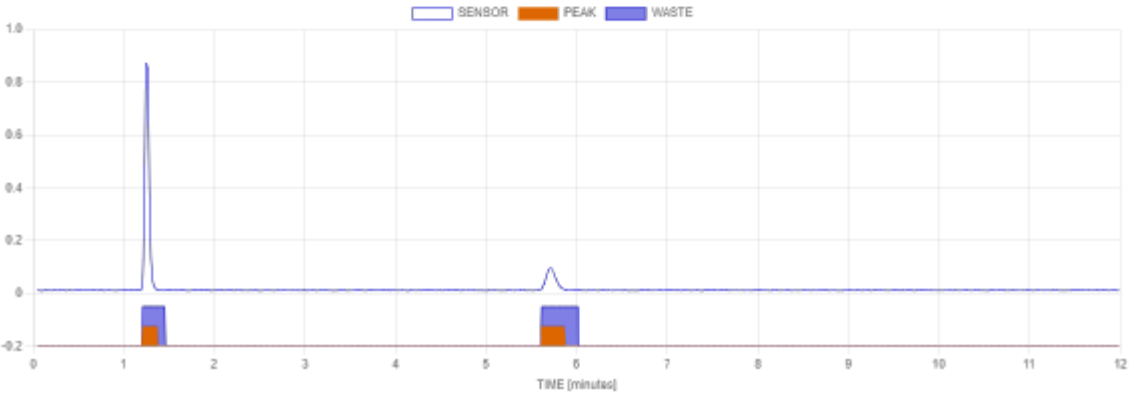


Figure 4: Sample Chromatogram report from ReSolve demonstrating the waste events and volumes for paracetamol and ibuprofen peaks. *(Injection: Sequence 2 Nuromol Sample - Run 70)*

5. Discussion

Using the Recycled mobile phase from Sequence 1 (Collection Stage) as the mobile phase for Sequence 2 (Evaluation Stage), the PMBrc observed no adverse impact on chromatography.

For both sequences, 94.5% of total mobile phase was recycled, with only 5.5% directed to the waste container. By reducing mobile phase consumption and subsequent hazardous waste generation a laboratory can expect:

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APPENDIX 1 – Chromatographic Metrics

Sequence 1 – Collection Stage

Paracetamol Chromatographic Indicators

Vial #	Name/Type	t_R (mins)	Area (mAU.s)	Tailing	Symmetry
1	Mobile Phase	n/a	n/a	n/a	n/a
2	Std - 99.5 µg/mL	1.21	1389.2	1.19	0.86
3	Std - 99.5 µg/mL	1.21	1386.1	1.15	0.87
4	Std - 99.5 µg/mL	1.21	1393.2	1.06	0.87
5	Std - 99.5 µg/mL	1.21	1388.6	1.10	0.88
6	Std - 99.5 µg/mL	1.21	1389.2	1.10	0.88
7	Nuromol Sample - Injection 1	1.21	1656.2	1.17	0.87
8	Nuromol Sample - Injection 2	1.21	1669.4	1.08	0.87
9	Nuromol Sample - Injection 3	1.21	1670.7	1.16	0.86
10	Nuromol Sample - Injection 4	1.21	1667.1	1.11	0.87
11	Nuromol Sample - Injection 5	1.21	1668.9	1.18	0.86
12	Nuromol Sample - Injection 6	1.21	1669.6	1.14	0.86
13	Nuromol Sample - Injection 7	1.21	1668.7	1.05	0.86
14	Nuromol Sample - Injection 8	1.21	1668.3	1.17	0.87
15	Nuromol Sample - Injection 9	1.21	1669.4	1.09	0.87
16	Nuromol Sample - Injection 10	1.21	1667.8	1.20	0.86
17	Std - 99.5 µg/mL	1.21	1391.2	1.15	0.87
18	Nuromol Sample - Injection 11	1.21	1667.5	1.11	0.87
19	Nuromol Sample - Injection 12	1.21	1673.3	1.11	0.87
20	Nuromol Sample - Injection 13	1.21	1677.4	1.14	0.86
21	Nuromol Sample - Injection 14	1.21	1669.9	1.05	0.87
22	Nuromol Sample - Injection 15	1.21	1681.1	1.17	0.87
23	Nuromol Sample - Injection 16	1.21	1688.5	1.16	0.87
24	Nuromol Sample - Injection 17	1.21	1677.9	1.08	0.87
25	Nuromol Sample - Injection 18	1.21	1671.9	1.16	0.86
26	Nuromol Sample - Injection 19	1.21	1670.2	1.11	0.87
27	Nuromol Sample - Injection 20	1.21	1668.7	1.19	0.86
28	Std - 99.5 µg/mL	1.21	1393.1	1.10	0.88
29	Nuromol Sample - Injection 21	1.21	1667.2	1.06	0.87

30	Nuromol Sample - Injection 22	1.21	1670.0	1.13	0.86
31	Nuromol Sample - Injection 23	1.21	1675.2	1.10	0.87
32	Nuromol Sample - Injection 24	1.21	1668.6	1.08	0.87
33	Nuromol Sample - Injection 25	1.21	1680.2	1.07	0.87
34	Nuromol Sample - Injection 26	1.21	1681.2	1.12	0.87
35	Nuromol Sample - Injection 27	1.21	1674.0	1.11	0.87
36	Nuromol Sample - Injection 28	1.21	1677.1	1.07	0.87
37	Nuromol Sample - Injection 29	1.21	1673.0	1.11	0.87
38	Nuromol Sample - Injection 30	1.21	1670.3	1.06	0.87
39	Std - 99.5 µg/mL	1.21	1391.9	1.04	0.87
40	Nuromol Sample - Injection 31	1.21	1670.1	1.12	0.87
41	Nuromol Sample - Injection 32	1.21	1670.5	1.20	0.86
42	Nuromol Sample - Injection 33	1.21	1668.9	1.20	0.86
43	Nuromol Sample - Injection 34	1.21	1675.2	1.11	0.87
44	Nuromol Sample - Injection 35	1.21	1671.9	1.14	0.86
45	Nuromol Sample - Injection 36	1.21	1671.2	1.14	0.86
46	Nuromol Sample - Injection 37	1.21	1671.6	1.05	0.87
47	Nuromol Sample - Injection 38	1.21	1670.5	1.17	0.86
48	Nuromol Sample - Injection 39	1.21	1670.3	1.12	0.87
49	Nuromol Sample - Injection 40	1.21	1670.0	1.08	0.87
50	Std - 99.5 µg/mL	1.21	1395.0	1.19	0.86
51	Nuromol Sample - Injection 41	1.21	1669.4	1.10	0.87
52	Nuromol Sample - Injection 42	1.21	1678.4	1.10	0.87
53	Nuromol Sample - Injection 43	1.21	1668.6	1.10	0.87
54	Nuromol Sample - Injection 44	1.21	1683.5	1.13	0.86
55	Nuromol Sample - Injection 45	1.21	1675.6	1.04	0.86
56	Nuromol Sample - Injection 46	1.21	1672.2	1.15	0.86
57	Nuromol Sample - Injection 47	1.21	1675.2	1.07	0.87
58	Nuromol Sample - Injection 48	1.21	1671.0	1.08	0.87
59	Nuromol Sample - Injection 49	1.21	1667.7	1.18	0.86
60	Nuromol Sample - Injection 50	1.21	1670.3	1.05	0.87
61	Std - 99.5 µg/mL	1.21	1392.6	1.05	0.87
62	Mobile Phase	n/a	n/a	n/a	n/a
63	Nuromol Sample - Injection 51	1.21	1671.4	1.13	0.86
64	Nuromol Sample - Injection 52	1.21	1673.3	1.20	0.86
65	Nuromol Sample - Injection 53	1.21	1671.5	1.15	0.86
66	Nuromol Sample - Injection 54	1.21	1675.1	1.15	0.86
67	Nuromol Sample - Injection 55	1.21	1670.2	1.06	0.87
68	Nuromol Sample - Injection 56	1.21	1668.2	1.10	0.87

69	Nuromol Sample - Injection 57	1.21	1668.1	1.13	0.86
70	Nuromol Sample - Injection 58	1.21	1667.2	1.13	0.86
71	Nuromol Sample - Injection 59	1.21	1670.9	1.06	0.87
72	Nuromol Sample - Injection 60	1.21	1669.5	1.10	0.87
73	Std - 99.5 µg/mL	1.21	1396.1	1.17	0.87
74	Nuromol Sample - Injection 61	1.21	1669.5	1.20	0.86
75	Nuromol Sample - Injection 62	1.21	1663.1	1.11	0.87
76	Nuromol Sample - Injection 63	1.21	1672.4	1.10	0.87
77	Nuromol Sample - Injection 64	1.21	1669.6	1.10	0.87
78	Nuromol Sample - Injection 65	1.21	1669.7	1.13	0.86
79	Nuromol Sample - Injection 66	1.21	1669.2	1.09	0.87
80	Nuromol Sample - Injection 67	1.21	1669.9	1.16	0.86
81	Nuromol Sample - Injection 68	1.21	1669.7	1.16	0.86
82	Nuromol Sample - Injection 69	1.21	1671.1	1.07	0.87
83	Nuromol Sample - Injection 70	1.21	1665.6	1.11	0.87
84	Std - 99.5 µg/mL	1.21	1391.4	1.14	0.86
85	Nuromol Sample - Injection 71	1.21	1665.0	1.16	0.86
86	Nuromol Sample - Injection 72	1.21	1667.5	1.20	0.86
87	Nuromol Sample - Injection 73	1.21	1666.2	1.12	0.87
88	Nuromol Sample - Injection 74	1.21	1665.6	1.12	0.87
89	Nuromol Sample - Injection 75	1.21	1665.5	1.11	0.87
90	Nuromol Sample - Injection 76	1.21	1669.2	1.14	0.86
91	Nuromol Sample - Injection 77	1.21	1665.9	1.14	0.86
92	Nuromol Sample - Injection 78	1.21	1670.1	1.13	0.86
93	Nuromol Sample - Injection 79	1.21	1670.9	1.09	0.87
94	Nuromol Sample - Injection 80	1.21	1666.4	1.12	0.87
95	Std - 99.5 µg/mL	1.21	1390.1	1.16	0.86
96	Mobile Phase	n/a	n/a	n/a	n/a

Sequence 1 – Collection Stage
Ibuprofen Chromatographic Indicators

Vial #	Name/Type	t_R (mins)	Area (mAU.s)	Tailing	Symmetry	Resolution EP
1	Mobile Phase	n/a	n/a	n/a	n/a	n/a
2	Std - 40.6 µg/mL	5.64	322.3	1.19	0.78	30.48
3	Std - 40.6 µg/mL	5.64	321.6	1.19	0.78	30.52
4	Std - 40.6 µg/mL	5.64	322.6	1.16	0.77	30.46
5	Std - 40.6 µg/mL	5.64	322.4	1.18	0.77	30.53
6	Std - 40.6 µg/mL	5.64	322.9	1.18	0.77	30.53
7	Nuromol Sample - Injection 1	5.63	373.4	1.19	0.75	30.38
8	Nuromol Sample - Injection 2	5.63	376.1	1.23	0.75	30.32
9	Nuromol Sample - Injection 3	5.63	376.6	1.19	0.75	30.38
10	Nuromol Sample - Injection 4	5.64	375.4	1.17	0.75	30.36
11	Nuromol Sample - Injection 5	5.64	376.2	1.21	0.75	30.33
12	Nuromol Sample - Injection 6	5.64	376.5	1.20	0.76	30.38
13	Nuromol Sample - Injection 7	5.64	376.0	1.24	0.75	30.23
14	Nuromol Sample - Injection 8	5.64	376.2	1.23	0.75	30.32
15	Nuromol Sample - Injection 9	5.64	376.0	1.20	0.75	30.33
16	Nuromol Sample - Injection 10	5.64	376.3	1.18	0.75	30.30
17	Std - 40.6 µg/mL	5.64	322.9	1.19	0.77	30.54
18	Nuromol Sample - Injection 11	5.64	375.8	1.23	0.75	30.35
19	Nuromol Sample - Injection 12	5.64	376.8	1.23	0.75	30.33
20	Nuromol Sample - Injection 13	5.64	376.4	1.25	0.75	30.32
21	Nuromol Sample - Injection 14	5.64	376.2	1.21	0.76	30.29
22	Nuromol Sample - Injection 15	5.64	376.7	1.19	0.75	30.34
23	Nuromol Sample - Injection 16	5.64	376.7	1.19	0.75	30.35
24	Nuromol Sample - Injection 17	5.64	376.5	1.23	0.75	30.28
25	Nuromol Sample - Injection 18	5.64	376.9	1.20	0.75	30.35
26	Nuromol Sample - Injection 19	5.64	376.4	1.19	0.75	30.37
27	Nuromol Sample - Injection 20	5.64	375.6	1.23	0.75	30.27
28	Std - 40.6 µg/mL	5.65	323.3	1.22	0.77	30.47
29	Nuromol Sample - Injection 21	5.64	375.4	1.18	0.75	30.29
30	Nuromol Sample - Injection 22	5.64	376.1	1.22	0.75	30.35
31	Nuromol Sample - Injection 23	5.64	376.3	1.19	0.75	30.35
32	Nuromol Sample - Injection 24	5.64	375.9	1.18	0.75	30.31
33	Nuromol Sample - Injection 25	5.64	377.1	1.19	0.75	30.32
34	Nuromol Sample - Injection 26	5.65	378.5	1.22	0.75	30.36

35	Nuromol Sample - Injection 27	5.65	377.0	1.22	0.75	30.34
36	Nuromol Sample - Injection 28	5.65	376.5	1.20	0.75	30.31
37	Nuromol Sample - Injection 29	5.65	376.0	1.22	0.75	30.34
38	Nuromol Sample - Injection 30	5.65	375.9	1.19	0.75	30.29
39	Std - 40.6 µg/mL	5.65	322.5	1.21	0.77	30.42
40	Nuromol Sample - Injection 31	5.65	375.7	1.22	0.75	30.34
41	Nuromol Sample - Injection 32	5.65	375.9	1.19	0.75	30.29
42	Nuromol Sample - Injection 33	5.65	375.8	1.18	0.75	30.28
43	Nuromol Sample - Injection 34	5.65	376.6	1.23	0.75	30.34
44	Nuromol Sample - Injection 35	5.65	375.9	1.25	0.75	30.29
45	Nuromol Sample - Injection 36	5.65	376.0	1.24	0.75	30.29
46	Nuromol Sample - Injection 37	5.65	376.0	1.21	0.75	30.25
47	Nuromol Sample - Injection 38	5.65	376.2	1.19	0.75	30.29
48	Nuromol Sample - Injection 39	5.65	375.9	1.24	0.75	30.28
49	Nuromol Sample - Injection 40	5.65	375.6	1.22	0.75	30.27
50	Std - 40.6 µg/mL	5.65	323.1	1.22	0.77	30.42
51	Nuromol Sample - Injection 41	5.65	375.4	1.23	0.75	30.33
52	Nuromol Sample - Injection 42	5.65	375.4	1.23	0.75	30.32
53	Nuromol Sample - Injection 43	5.65	375.5	1.23	0.75	30.29
54	Nuromol Sample - Injection 44	5.65	376.6	1.24	0.75	30.28
55	Nuromol Sample - Injection 45	5.65	375.9	1.21	0.75	30.22
56	Nuromol Sample - Injection 46	5.65	376.0	1.19	0.75	30.35
57	Nuromol Sample - Injection 47	5.65	376.0	1.23	0.75	30.27
58	Nuromol Sample - Injection 48	5.65	375.8	1.24	0.75	30.25
59	Nuromol Sample - Injection 49	5.65	375.7	1.22	0.75	30.27
60	Nuromol Sample - Injection 50	5.65	375.7	1.23	0.75	30.23
61	Std - 40.6 µg/mL	5.65	322.3	1.19	0.77	30.42
62	Mobile Phase	n/a	n/a	n/a	n/a	n/a
63	Nuromol Sample - Injection 51	5.65	375.6	1.20	0.75	30.33
64	Nuromol Sample - Injection 52	5.65	376.1	1.24	0.75	30.22
65	Nuromol Sample - Injection 53	5.65	375.7	1.22	0.75	30.31
66	Nuromol Sample - Injection 54	5.65	376.4	1.21	0.76	30.32
67	Nuromol Sample - Injection 55	5.65	375.6	1.18	0.75	30.25
68	Nuromol Sample - Injection 56	5.65	375.4	1.20	0.75	30.31
69	Nuromol Sample - Injection 57	5.65	375.1	1.21	0.75	30.33
70	Nuromol Sample - Injection 58	5.65	374.5	1.21	0.75	30.33
71	Nuromol Sample - Injection 59	5.65	375.8	1.21	0.75	30.25
72	Nuromol Sample - Injection 60	5.65	375.5	1.23	0.75	30.27
73	Std - 40.6 µg/mL	5.66	323.0	1.20	0.77	30.46

74	Nuromol Sample - Injection 61	5.65	375.3	1.18	0.75	30.25
75	Nuromol Sample - Injection 62	5.65	373.6	1.21	0.75	30.30
76	Nuromol Sample - Injection 63	5.65	376.2	1.20	0.75	30.28
77	Nuromol Sample - Injection 64	5.65	375.5	1.20	0.75	30.29
78	Nuromol Sample - Injection 65	5.65	374.9	1.22	0.75	30.30
79	Nuromol Sample - Injection 66	5.65	374.8	1.20	0.75	30.28
80	Nuromol Sample - Injection 67	5.65	375.2	1.24	0.75	30.27
81	Nuromol Sample - Injection 68	5.65	375.2	1.24	0.75	30.27
82	Nuromol Sample - Injection 69	5.65	375.4	1.20	0.75	30.28
83	Nuromol Sample - Injection 70	5.65	374.4	1.22	0.75	30.29
84	Std - 40.6 µg/mL	5.66	322.3	1.18	0.77	30.49
85	Nuromol Sample - Injection 71	5.66	374.8	1.24	0.75	30.29
86	Nuromol Sample - Injection 72	5.66	375.0	1.19	0.75	30.29
87	Nuromol Sample - Injection 73	5.66	375.5	1.25	0.75	30.29
88	Nuromol Sample - Injection 74	5.66	375.3	1.25	0.75	30.29
89	Nuromol Sample - Injection 75	5.66	374.8	1.18	0.75	30.33
90	Nuromol Sample - Injection 76	5.66	375.6	1.19	0.75	30.35
91	Nuromol Sample - Injection 77	5.66	374.9	1.20	0.75	30.35
92	Nuromol Sample - Injection 78	5.66	375.8	1.19	0.75	30.34
93	Nuromol Sample - Injection 79	5.66	375.8	1.25	0.75	30.27
94	Nuromol Sample - Injection 80	5.66	375.2	1.20	0.75	30.33
95	Std - 40.6 µg/mL	5.67	322.0	1.17	0.77	30.51
96	Mobile Phase	n/a	n/a	n/a	n/a	n/a

Sequence 2 – Evaluation Stage

Paracetamol Chromatographic Indicators

Vial #	Name/Type	t_R (mins)	Area (mAU.s)	Tailing	Symmetry
1	Recycled Mobile Phase	n/a	n/a	n/a	n/a
2	Std - 99.5 µg/mL	1.21	1394.3	1.16	0.87
3	Std - 99.5 µg/mL	1.21	1398.4	1.04	0.87
4	Std - 99.5 µg/mL	1.21	1404.0	1.11	0.87
5	Std - 99.5 µg/mL	1.21	1398.0	1.18	0.86
6	Std - 99.5 µg/mL	1.21	1400.0	1.13	0.86
7	Nuromol Sample - Injection 1	1.21	1667.3	1.04	0.86
8	Nuromol Sample - Injection 2	1.21	1671.6	1.12	0.87
9	Nuromol Sample - Injection 3	1.21	1672.0	1.07	0.87
10	Nuromol Sample - Injection 4	1.21	1670.8	1.11	0.87
11	Nuromol Sample - Injection 5	1.21	1672.3	1.11	0.87
12	Nuromol Sample - Injection 6	1.21	1673.5	1.19	0.86
13	Nuromol Sample - Injection 7	1.21	1671.9	1.17	0.86
14	Nuromol Sample - Injection 8	1.21	1669.0	1.17	0.86
15	Nuromol Sample - Injection 9	1.21	1670.9	1.17	0.86
16	Nuromol Sample - Injection 10	1.21	1672.0	1.16	0.86
17	Std - 99.5 µg/mL	1.21	1392.9	1.08	0.87
18	Nuromol Sample - Injection 11	1.21	1663.6	1.07	0.87
19	Nuromol Sample - Injection 12	1.21	1665.2	1.06	0.87
20	Nuromol Sample - Injection 13	1.21	1664.5	1.14	0.86
21	Nuromol Sample - Injection 14	1.21	1669.8	1.11	0.87
22	Nuromol Sample - Injection 15	1.21	1669.6	1.21	0.86
23	Nuromol Sample - Injection 16	1.21	1670.9	1.09	0.87
24	Nuromol Sample - Injection 17	1.21	1670.7	1.08	0.87
25	Nuromol Sample - Injection 18	1.21	1669.8	1.15	0.86
26	Nuromol Sample - Injection 19	1.21	1669.3	1.15	0.86
27	Nuromol Sample - Injection 20	1.21	1671.2	1.06	0.87
28	Std - 99.5 µg/mL	1.21	1397.8	1.14	0.86
29	Nuromol Sample - Injection 21	1.21	1671.7	1.14	0.86
30	Nuromol Sample - Injection 22	1.21	1667.3	1.09	0.87
31	Nuromol Sample - Injection 23	1.21	1669.9	1.16	0.86
32	Nuromol Sample - Injection 24	1.21	1671.4	1.12	0.87
33	Nuromol Sample - Injection 25	1.21	1673.6	1.12	0.87
34	Nuromol Sample - Injection 26	1.21	1675.5	1.14	0.86

35	Nuromol Sample - Injection 27	1.21	1674.8	1.06	0.87
36	Nuromol Sample - Injection 28	1.21	1678.0	1.05	0.86
37	Nuromol Sample - Injection 29	1.21	1673.0	1.05	0.86
38	Nuromol Sample - Injection 30	1.21	1673.6	1.17	0.86
39	Std - 99.5 µg/mL	1.21	1409.6	1.13	0.86
40	Nuromol Sample - Injection 31	1.21	1672.0	1.08	0.87
41	Nuromol Sample - Injection 32	1.21	1669.6	1.19	0.86
42	Nuromol Sample - Injection 33	1.21	1673.9	1.06	0.87
43	Nuromol Sample - Injection 34	1.21	1679.8	1.14	0.86
44	Nuromol Sample - Injection 35	1.21	1671.8	1.05	0.86
45	Nuromol Sample - Injection 36	1.21	1677.8	1.16	0.86
46	Nuromol Sample - Injection 37	1.21	1681.8	1.20	0.86
47	Nuromol Sample - Injection 38	1.21	1687.9	1.08	0.87
48	Nuromol Sample - Injection 39	1.21	1671.7	1.07	0.87
49	Nuromol Sample - Injection 40	1.21	1674.7	1.07	0.87
50	Std - 99.5 µg/mL	1.21	1399.0	1.13	0.86
51	Nuromol Sample - Injection 41	1.21	1673.6	1.06	0.87
52	Nuromol Sample - Injection 42	1.21	1693.6	1.09	0.87
53	Nuromol Sample - Injection 43	1.21	1680.4	1.09	0.87
54	Nuromol Sample - Injection 44	1.21	1679.1	1.10	0.87
55	Nuromol Sample - Injection 45	1.21	1673.8	1.10	0.87
56	Nuromol Sample - Injection 46	1.21	1672.9	1.07	0.87
57	Nuromol Sample - Injection 47	1.21	1670.5	1.09	0.87
58	Nuromol Sample - Injection 48	1.21	1672.4	1.10	0.87
59	Nuromol Sample - Injection 49	1.21	1673.5	1.08	0.87
60	Nuromol Sample - Injection 50	1.21	1672.7	1.16	0.86
61	Std - 99.5 µg/mL	1.21	1411.5	1.20	0.86
62	Recycled Mobile Phase	n/a	n/a	n/a	n/a
63	Nuromol Sample - Injection 51	1.21	1667.3	1.18	0.86
64	Nuromol Sample - Injection 52	1.21	1673.0	1.15	0.86
65	Nuromol Sample - Injection 53	1.21	1672.6	1.17	0.86
66	Nuromol Sample - Injection 54	1.21	1680.7	1.13	0.86
67	Nuromol Sample - Injection 55	1.21	1667.9	1.20	0.86
68	Nuromol Sample - Injection 56	1.21	1668.6	1.13	0.86
69	Nuromol Sample - Injection 57	1.21	1667.8	1.15	0.86
70	Nuromol Sample - Injection 58	1.21	1672.8	1.08	0.87
71	Nuromol Sample - Injection 59	1.21	1670.0	1.18	0.86
72	Nuromol Sample - Injection 60	1.21	1672.0	1.14	0.86
73	Std - 99.5 µg/mL	1.21	1415.5	1.14	0.86

74	Nuromol Sample - Injection 61	1.21	1669.8	1.17	0.86
75	Nuromol Sample - Injection 62	1.21	1669.9	1.10	0.87
76	Nuromol Sample - Injection 63	1.21	1678.8	1.09	0.87
77	Nuromol Sample - Injection 64	1.21	1678.1	1.09	0.87
78	Nuromol Sample - Injection 65	1.21	1677.4	1.20	0.86
79	Nuromol Sample - Injection 66	1.21	1677.3	1.19	0.86
80	Nuromol Sample - Injection 67	1.21	1685.0	1.09	0.87
81	Nuromol Sample - Injection 68	1.21	1681.5	1.12	0.87
82	Nuromol Sample - Injection 69	1.21	1672.6	1.13	0.86
83	Nuromol Sample - Injection 70	1.21	1668.9	1.05	0.87
84	Std - 99.5 µg/mL	1.21	1396.9	1.10	0.87
85	Nuromol Sample - Injection 71	1.21	1668.8	1.16	0.86
86	Nuromol Sample - Injection 72	1.21	1667.3	1.08	0.87
87	Nuromol Sample - Injection 73	1.21	1670.0	1.19	0.86
88	Nuromol Sample - Injection 74	1.21	1675.6	1.11	0.87
89	Nuromol Sample - Injection 75	1.21	1680.6	1.19	0.86
90	Nuromol Sample - Injection 76	1.21	1680.5	1.12	0.87
91	Nuromol Sample - Injection 77	1.21	1674.1	1.05	0.86
92	Nuromol Sample - Injection 78	1.21	1675.4	1.04	0.86
93	Nuromol Sample - Injection 79	1.21	1667.4	1.08	0.87
94	Nuromol Sample - Injection 80	1.21	1666.7	1.08	0.87
95	Std - 99.5 µg/mL	1.21	1395.1	1.13	0.86
96	Recycled Mobile Phase	n/a	n/a	n/a	n/a

Sequence 2 – Evaluation Stage

Ibuprofen Chromatographic Indicators

Vial #	Name/Type	t_R (mins)	Area (mAU.s)	Tailing	Symmetry	Resolution EP
1	Recycled Mobile Phase	n/a	n/a	n/a	n/a	n/a
2	Std - 40.6 µg/mL	5.70	322.2	1.20	0.77	30.54
3	Std - 40.6 µg/mL	5.69	322.0	1.16	0.77	30.43
4	Std - 40.6 µg/mL	5.69	324.2	1.16	0.77	30.50
5	Std - 40.6 µg/mL	5.69	323.2	1.17	0.77	30.46
6	Std - 40.6 µg/mL	5.69	323.3	1.20	0.77	30.51
7	Nuromol Sample - Injection 1	5.68	374.3	1.19	0.75	30.23
8	Nuromol Sample - Injection 2	5.68	376.0	1.20	0.75	30.30
9	Nuromol Sample - Injection 3	5.68	375.5	1.25	0.75	30.26
10	Nuromol Sample - Injection 4	5.68	375.9	1.19	0.75	30.33
11	Nuromol Sample - Injection 5	5.68	375.8	1.24	0.75	30.29
12	Nuromol Sample - Injection 6	5.68	376.1	1.22	0.75	30.30
13	Nuromol Sample - Injection 7	5.67	376.0	1.19	0.75	30.32
14	Nuromol Sample - Injection 8	5.67	375.9	1.18	0.75	30.31
15	Nuromol Sample - Injection 9	5.67	375.6	1.18	0.75	30.30
16	Nuromol Sample - Injection 10	5.67	375.9	1.24	0.75	30.28
17	Std - 40.6 µg/mL	5.68	322.9	1.16	0.77	30.50
18	Nuromol Sample - Injection 11	5.67	374.1	1.19	0.75	30.30
19	Nuromol Sample - Injection 12	5.67	375.0	1.18	0.75	30.28
20	Nuromol Sample - Injection 13	5.67	374.3	1.22	0.75	30.34
21	Nuromol Sample - Injection 14	5.67	375.3	1.21	0.75	30.35
22	Nuromol Sample - Injection 15	5.67	375.3	1.25	0.75	30.24
23	Nuromol Sample - Injection 16	5.67	375.7	1.19	0.75	30.32
24	Nuromol Sample - Injection 17	5.67	375.6	1.18	0.75	30.30
25	Nuromol Sample - Injection 18	5.67	375.4	1.20	0.75	30.36
26	Nuromol Sample - Injection 19	5.67	375.3	1.21	0.75	30.36
27	Nuromol Sample - Injection 20	5.67	375.4	1.24	0.75	30.27
28	Std - 40.6 µg/mL	5.67	323.8	1.22	0.77	30.55
29	Nuromol Sample - Injection 21	5.67	375.5	1.19	0.75	30.38
30	Nuromol Sample - Injection 22	5.67	374.5	1.25	0.75	30.30
31	Nuromol Sample - Injection 23	5.67	374.9	1.20	0.75	30.36
32	Nuromol Sample - Injection 24	5.67	375.9	1.18	0.75	30.34
33	Nuromol Sample - Injection 25	5.67	376.4	1.25	0.75	30.30
34	Nuromol Sample - Injection 26	5.67	375.8	1.18	0.75	30.36

35	Nuromol Sample - Injection 27	5.67	375.0	1.22	0.75	30.30
36	Nuromol Sample - Injection 28	5.67	376.1	1.21	0.75	30.26
37	Nuromol Sample - Injection 29	5.67	375.3	1.20	0.75	30.25
38	Nuromol Sample - Injection 30	5.67	375.4	1.18	0.75	30.35
39	Std - 40.6 µg/mL	5.67	324.7	1.18	0.77	30.50
40	Nuromol Sample - Injection 31	5.67	376.0	1.22	0.75	30.29
41	Nuromol Sample - Injection 32	5.67	374.3	1.19	0.75	30.32
42	Nuromol Sample - Injection 33	5.67	375.4	1.20	0.75	30.29
43	Nuromol Sample - Injection 34	5.67	376.1	1.24	0.75	30.30
44	Nuromol Sample - Injection 35	5.67	374.9	1.19	0.75	30.24
45	Nuromol Sample - Injection 36	5.66	375.8	1.24	0.75	30.28
46	Nuromol Sample - Injection 37	5.66	375.7	1.18	0.75	30.26
47	Nuromol Sample - Injection 38	5.66	375.1	1.20	0.75	30.29
48	Nuromol Sample - Injection 39	5.66	375.0	1.19	0.75	30.28
49	Nuromol Sample - Injection 40	5.66	375.9	1.19	0.75	30.26
50	Std - 40.6 µg/mL	5.67	323.5	1.17	0.77	30.50
51	Nuromol Sample - Injection 41	5.66	376.0	1.18	0.75	30.20
52	Nuromol Sample - Injection 42	5.66	376.0	1.20	0.76	30.29
53	Nuromol Sample - Injection 43	5.66	376.9	1.19	0.75	30.28
54	Nuromol Sample - Injection 44	5.66	375.9	1.17	0.75	30.29
55	Nuromol Sample - Injection 45	5.66	375.2	1.24	0.75	30.24
56	Nuromol Sample - Injection 46	5.66	375.9	1.23	0.75	30.19
57	Nuromol Sample - Injection 47	5.66	375.2	1.24	0.75	30.22
58	Nuromol Sample - Injection 48	5.66	375.8	1.24	0.76	30.20
59	Nuromol Sample - Injection 49	5.66	375.2	1.23	0.75	30.24
60	Nuromol Sample - Injection 50	5.66	375.0	1.20	0.76	30.27
61	Std - 40.6 µg/mL	5.66	322.3	1.16	0.77	30.43
62	Recycled Mobile Phase	n/a	n/a	n/a	n/a	n/a
63	Nuromol Sample - Injection 51	5.66	374.3	1.22	0.75	30.23
64	Nuromol Sample - Injection 52	5.66	375.2	1.19	0.75	30.30
65	Nuromol Sample - Injection 53	5.66	374.9	1.21	0.75	30.29
66	Nuromol Sample - Injection 54	5.66	376.3	1.19	0.76	30.29
67	Nuromol Sample - Injection 55	5.66	374.2	1.22	0.75	30.23
68	Nuromol Sample - Injection 56	5.66	374.2	1.21	0.75	30.28
69	Nuromol Sample - Injection 57	5.66	373.9	1.23	0.75	30.29
70	Nuromol Sample - Injection 58	5.66	374.9	1.18	0.75	30.26
71	Nuromol Sample - Injection 59	5.66	375.1	1.22	0.75	30.23
72	Nuromol Sample - Injection 60	5.66	374.1	1.21	0.75	30.31
73	Std - 40.6 µg/mL	5.67	322.7	1.16	0.77	30.47

74	Nuromol Sample - Injection 61	5.66	374.4	1.23	0.76	30.24
75	Nuromol Sample - Injection 62	5.66	373.7	1.19	0.76	30.27
76	Nuromol Sample - Injection 63	5.66	374.6	1.19	0.75	30.26
77	Nuromol Sample - Injection 64	5.66	374.2	1.19	0.76	30.26
78	Nuromol Sample - Injection 65	5.66	374.3	1.25	0.75	30.19
79	Nuromol Sample - Injection 66	5.67	374.6	1.20	0.76	30.23
80	Nuromol Sample - Injection 67	5.67	374.2	1.23	0.75	30.23
81	Nuromol Sample - Injection 68	5.66	374.5	1.21	0.75	30.28
82	Nuromol Sample - Injection 69	5.66	374.7	1.21	0.75	30.26
83	Nuromol Sample - Injection 70	5.66	374.3	1.18	0.75	30.17
84	Std - 40.6 µg/mL	5.67	322.3	1.16	0.77	30.43
85	Nuromol Sample - Injection 71	5.66	374.5	1.24	0.76	30.19
86	Nuromol Sample - Injection 72	5.67	373.7	1.21	0.75	30.23
87	Nuromol Sample - Injection 73	5.66	373.9	1.19	0.75	30.25
88	Nuromol Sample - Injection 74	5.67	375.1	1.23	0.75	30.23
89	Nuromol Sample - Injection 75	5.67	374.7	1.19	0.76	30.24
90	Nuromol Sample - Injection 76	5.66	374.7	1.22	0.75	30.28
91	Nuromol Sample - Injection 77	5.67	374.5	1.22	0.75	30.20
92	Nuromol Sample - Injection 78	5.67	374.8	1.22	0.75	30.19
93	Nuromol Sample - Injection 79	5.67	373.8	1.23	0.76	30.25
94	Nuromol Sample - Injection 80	5.67	373.8	1.24	0.76	30.21
95	Std - 40.6 µg/mL	5.67	322.8	1.20	0.77	30.48
96	Recycled Mobile Phase	n/a	n/a	n/a	n/a	n/a